FYP-Modules

1. Project Identification & Metadata

- **Working Acronym / Short Title:** LedgerIQ Studio
- **Official Academic Title:** LedgerIQ: Autonomous Neuro-Symbolic Engine for Financial Document Parsing, Reconciliation, and Statutory Audit Verification
- **Alternative Title:** Multimodal Document Understanding and Deterministic Verification System for Financial Statement Compilation
- **Declared Academic Track:** Stream A (Research-Derived System Engineering Track, validated against enterprise accounting workflows)
- **Accreditation Standard:** Seoul Accord Complex Computing Problem (CCP) Framework (Graduate Attributes GA-2: Abstraction, GA-4: Solution Design & Evaluation)
- **Target Jurisdiction & Accounting Standard:** International Financial Reporting Standards (IFRS) and the Companies Act 2017 (Fourth and Fifth Schedules)
- **Target Hardware Execution Profile:** On-premises consumer edge GPU (NVIDIA RTX 3050 / RTX 3070, 4 GB to 8 GB VRAM) with zero external cloud API dependencies

2. Executive Summary & Problem Formulation

Operational Friction in the Accounting & Audit Market

In corporate accounting, professional bookkeeping, and statutory financial audit firms, junior accountants and audit trainees spend 40% to 60% of their billable hours manually transcribing unstandardized, multi-page financial documents (scanned trial balances, general ledgers, bank statements, tax challans, and vendor invoices) into spreadsheets. Beyond basic transcription, accounting workflows require performing year-end adjusting journal entries (depreciation, prepayments, accruals), compiling statutory disclosure notes, executing multi-source bank reconciliations, verifying regional tax withholdings, and testing journal entries for fraud and anomalies.

Why Existing Solutions Fail

- **Commercial Document Utilities (Nanonets, Adobe PDF, Docsumo):** Process document pages as disconnected collections of bounding boxes. They regularly drop negative parenthetical notations (converting negative liabilities to positive numbers), misread accounting zero-dashes as missing cells, and break multi-page continuing tables into disjointed fragments.
- **Frontier Large Language Models (GPT-4o, Claude 3.5):** Autoregressive language models generate tokens probabilistically rather than symbolically. When transcribing dense numerical columns, they are susceptible to single-digit hallucinations or dropped values that invalidate balance sheet invariants ($\text{Debits} \not\equiv \text{Credits}$).
- **Basic Scripting / CRUD Applications:** While clean, single-page Excel files can be processed with basic scripts, real-world accounting is dominated by unadjusted balances, missing statutory note schedules, fuzzy bank transaction descriptions, and complex withholding tax structures.

The Neuro-Symbolic Core Paradigm

LedgerIQ resolves the tension between probabilistic visual perception and deterministic accounting rules by coupling two distinct computing engines into a closed feedback loop:

1. **Neural Visual Perception (The "Eye"):** Lightweight, 4-bit quantized open-weight vision-language models (MonkeyOCRv2 0.7B / OvisOCR2 0.8B) extract text, bounding boxes, and hierarchical table structures from messy documents without sending data to public cloud APIs.
2. **Symbolic Mathematical Verification (The "Auditor"):** Deterministic Linear Programming (LP) and Constraint Satisfaction Problem (CSP) solvers enforce non-negotiable accounting conservation laws:

$$\sum \text{Debits} - \sum \text{Credits} = 0$$
$$\sum_{i \in \text{SubAccounts}} \text{Balance}_i - \text{ParentAccountTotal} = 0$$
$$\text{Total Assets} - (\text{Total Liabilities} + \text{Total Equity}) = 0$$

3. **Closed-Loop Error Recovery:** If an arithmetic check fails ($\Delta \ne 0.00$), the symbolic engine identifies the exact row or cell causing the imbalance, extracts its spatial bounding box, and routes a high-resolution localized visual crop back for character-level disambiguation or interactive audit verification.

3. Comprehensive 9-Module System Specification

Module Classification Matrix

Module Identifier & Title	Complexity Type	Algorithmic Approach	Phased Delivery
Module 1: Intelligent Chart-of-Accounts (COA) Normalizer & Statutory Grouping	**[CCP Module]**	Sentence-transformer semantic embeddings + balance polarity constraint rules	Semester 1 (FYP-1 Core)
Module 2: Year-End Adjusting Journal Entry (AJE) & Working Paper Engine	**[CCP Module]**	Directed Acyclic Graph (DAG) state-space dependency modeling	Semester 1 (FYP-1 Core)
Module 3: Statutory Financial Statement & Notes Synthesizer (Companies Act 2017)	**[CCP Module]**	Hierarchical financial statement compiler enforcing Fourth/Fifth Schedule reporting	Semester 1 (FYP-1 Core)

Module 4: Automated Bank Reconciliation Statement (BRS) Engine	**[CCP Module]**	Fuzzy Bipartite Graph Matching solved via Mixed-Integer Linear Programming (MILP)	Semester 2 (FYP-2 Extension)
Module 5: Sales Tax & Withholding Tax (WHT / FBR) Reconciliation Matrix	**[CCP Module]**	Deterministic expert system evaluating multi-tier statutory tax rate matrices	Semester 2 (FYP-2 Extension)
Module 6: Non-Linear 3-Way Reconciliation & Invoice Matching Engine	**[CCP Module]**	Bipartite graph optimization under fractional fulfillment and unit-conversion constraints	Semester 2 (FYP-2 Extension)
Module 7: Lightweight Edge OCR & Receipt/Challan Digitizer	**[CCP Module]**	4-bit quantized vision-language model (0.7B/0.8B) running in under 2.2 GB VRAM	Semester 1 (FYP-1 Core)
Module 8: Audit Analytical Procedures & Anomaly Detector (ISA 520 / 240)	**[CCP Module]**	Benford's Law distribution analysis paired with Isolation Forest anomaly detection	Semester 2 (FYP-2 Extension)
Module 9: Dynamic OpenXML Excel Workpaper Pack Exporter	**[Support Module]**	Programmatic OpenXML compiler writing dynamic calculation formulas (=SUM(...))	Semester 1 (FYP-1 Core)

Detailed Module Profiles

Module 1: Intelligent Chart-of-Accounts (COA) Normalizer & Statutory Grouping

- **Classification:** [CCP Module - Complex Computing Problem]

- **Operational Pain Point:** Every corporate entity maintains a custom, idiosyncratic Chart of Accounts (e.g., Acc-102 Petty Cash HQ, 1020-Cash in Hand, Cash - Factory). Accountants spend hours manually re-classifying hundreds of client-specific accounts into standardized financial statement buckets.
- **Input Data:** Raw unstandardized Trial Balance files in .xlsx, .csv, or scanned .pdf format.
- **Algorithmic Workflow:**
    - Reads account descriptions and generates semantic text embeddings using a domain-adapted sentence transformer (all-MiniLM-L6-v2).
    - Evaluates account balance polarity (e.g., if a cash account shows a credit balance, the algorithm shifts it from "Current Assets" to "Current Liabilities" as a bank overdraft).
    - Maps each account to standardized IFRS/GAAP nodes and statutory reporting lines under the Companies Act 2017.
- **Required Output:** Normalized trial balance where every account is tagged with its parent category, financial statement line code, and reporting schedule identifier.
- **Application Layout Placement:** Middle financial grid (Taxonomy Mapping Drawer).
- **Seoul Accord Characteristics:** Characteristics 2 & 4 (Depth of analysis, unfamiliar domain issues).

Module 2: Year-End Adjusting Journal Entry (AJE) & Working Paper Engine

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Raw trial balances extracted from accounting software are unadjusted. Preparing financial statements requires calculating period-end adjustments—such as fixed asset depreciation schedules, prepayment amortizations, expense accruals, and bad-debt allowances. Performing these calculations across disconnected Excel tabs leads to broken references and formula errors.
- **Input Data:** Fixed asset register schedules, prepayment contract terms, accrual estimates, and the unadjusted trial balance.
- **Algorithmic Workflow:**
    - Computes depreciation schedules (Straight-Line, Reducing Balance) and prepayment amortizations across reporting periods.
    - Generates balanced double-entry adjusting entries ($\text{Debit Expense} / \text{Credit Accumulated Depreciation}$).
    - Propagates adjustments into the unadjusted ledger across a Directed Acyclic Graph (DAG), recalculating the Adjusted Working Trial Balance dynamically.
- **Required Output:** Complete Adjusting Journal Entry log and a balanced Adjusted Working Trial Balance with complete mathematical audit trails.
- **Application Layout Placement:** Middle financial grid (Adjustments column) and Right Inspector Panel.
- **Seoul Accord Characteristics:** Characteristics 1 & 8 (Conflicting constraints, system interdependence).

Module 3: Statutory Financial Statement & Numbered Notes Synthesizer (Companies Act 2017)

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Compiling the face of the Balance Sheet and Profit & Loss statement represents only a fraction of the statutory reporting workload. Under the

Fourth Schedule (for listed companies) and Fifth Schedule (for non-listed companies) of the Companies Act 2017, every balance sheet line item must reference a mandatory numbered explanatory note (e.g., *Note 14: Fixed Assets Schedule*, *Note 20: Trade Debts Breakup*, *Note 28: Contingencies and Commitments*). Drafting these notes manually takes days of labor.

- **Input Data:** Adjusted Working Trial Balance and underlying sub-ledger schedule tables.
- **Algorithmic Workflow:**
    - Aggregates sub-ledger accounts into primary statutory statements (Balance Sheet, Profit & Loss, Statement of Changes in Equity, Cash Flows).
    - Synthesizes the statutory notes hierarchy, assigning sequential note numbers and linking cross-reference identifiers on the face of the financial statements.
    - Compiles multi-column disclosure tables (cost, additions, disposals, depreciation, and net book value).
- **Required Output:** Complete statutory financial statement pack with fully cross-referenced numbered notes complying with the Fourth and Fifth Schedules.
- **Application Layout Placement:** Main workspace (Primary Statement Tabs and Numbered Notes Navigation Tree).
- **Seoul Accord Characteristics:** Characteristics 3 & 7 (Depth of domain knowledge, significant legal and statutory consequences).

Module 4: Automated Bank Reconciliation Statement (BRS) Engine

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Month-end bank reconciliation requires accountants to manually compare external bank statement PDFs (containing truncated descriptions like CHQ DEP 88123-PKR or FT-IBFT-00291) against internal cash book ledgers, searching for unpresented checks, uncleared deposits, and direct bank charges.
- **Input Data:** Bank statement PDF/CSV files and internal cash book/bank ledger exports.
- **Algorithmic Workflow:**
    - Formulates reconciliation as a Fuzzy Bipartite Graph Matching problem solved via Mixed-Integer Linear Programming (MILP).
    - Evaluates date-window tolerances ($\pm 5\text{ days}$ for check clearing), exact and split amount matching, and string edit-distance across transaction descriptions.
    - Classifies unmatched transactions into Unpresented Checks, Uncleared Deposits, or Direct Bank Deductions.
- **Required Output:** Standardized Bank Reconciliation Statement (BRS) calculating the exact bridge between bank and cash book balances, with categorized schedules of outstanding items.
- **Application Layout Placement:** Bank Reconciliation Studio view within the desktop workspace.
- **Seoul Accord Characteristics:** Characteristics 1, 2 & 5 (Conflicting requirements, no obvious solution, beyond standard practice).

Module 5: Sales Tax & Withholding Tax (WHT / FBR) Reconciliation Matrix

- **Classification:** [CCP Module - Complex Computing Problem]

- **Operational Pain Point:** Companies must deduct withholding taxes on vendor disbursements and file periodic sales tax returns (FBR Annex-C). Manual cross-checking of invoices against changing statutory tax rates leads to miscalculations, non-compliance notices, and monetary fines.
- **Input Data:** Line-item purchase registers, vendor payment vouchers, and regional tax rate configuration tables.
- **Algorithmic Workflow:**
    - Parses product and service classifications from extracted invoice line items.
    - Evaluates transactions against statutory tax matrices (standard vs. reduced sales tax, withholding rates on goods vs. services).
    - Compares calculated withholding amounts against ledger postings and flags under-deductions or improper tax exemptions.
- **Required Output:** Withholding Tax Reconciliation Report and Sales Tax Input Validation Schedule highlighting tax compliance variances.
- **Application Layout Placement:** Tax Compliance & WHT Validator tab in the statement compilation screen.
- **Seoul Accord Characteristics:** Characteristics 3 & 7 (Depth of domain knowledge, significant tax and legal consequences).

Module 6: Non-Linear 3-Way Reconciliation & Invoice Matching Engine

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Accounts Payable departments manually cross-reference supplier bills against Purchase Orders (POs) and Goods Received Notes (GRNs). Real-world procurement rarely matches 1-to-1 due to partial deliveries, line-item bundling, differing units of measure (e.g., ordered in cartons, invoiced in units), and discounts.
- **Input Data:** Scanned or digital Purchase Orders, Goods Received Notes, and Supplier Invoices.
- **Algorithmic Workflow:**
    - Parses item descriptions, quantities, unit prices, and line totals from all three documents.
    - Executes a bipartite graph matching optimization algorithm to pair invoice line items with corresponding PO and GRN entries under fractional fulfillment constraints.
    - Computes unit price variances, quantity variances, and tax differences.
- **Required Output:** 3-Way Reconciliation Audit Matrix with visual variance indicators (Green = Full Match, Yellow = Partial Delivery, Red = Price/Quantity Overcharge).
- **Application Layout Placement:** 3-Way Matching Studio displaying a multi-column document alignment canvas.
- **Seoul Accord Characteristics:** Characteristics 1, 2 & 8 (Conflicting requirements, depth of analysis, system interdependence).

Module 7: Lightweight Edge OCR & Receipt/Challan Digitizer

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Petty cash receipts, tax payment challans, and physical supplier bills arrive as noisy paper scans or mobile photos. Uploading client documents to public cloud AI APIs violates enterprise data privacy mandates (SOX, GDPR), while consumer hardware lacks memory for large vision models.

- **Input Data:** Mobile camera captures, scanned paper receipts, and tax payment challan images (150 to 300 DPI).
- **Algorithmic Workflow:**
    - Processes image slices through a 4-bit quantized open-weight vision model (MonkeyOCRv2 0.7B / OvisOCR2 0.8B) running locally in under 2.2 GB of GPU VRAM.
    - Extracts dates, invoice numbers, line items, and totals along with pixel-level spatial bounding boxes.
    - Runs an automated post-processing arithmetic check confirming that extracted line sums equal the extracted invoice total.
- **Required Output:** Normalized JSON/tabular records with spatial coordinate mappings, ready for ingestion into tax and reconciliation modules.
- **Application Layout Placement:** Document Ingestion & Scanner Pre-Flight view.
- **Seoul Accord Characteristics:** Characteristics 1 & 5 (Conflicting constraints, resource-constrained edge execution).

Module 8: Audit Analytical Procedures & Anomaly Detector (ISA 520 / ISA 240)

- **Classification:** [CCP Module - Complex Computing Problem]
- **Operational Pain Point:** Under International Standards on Auditing (ISA 520: Analytical Procedures and ISA 240: Fraud in Financial Statements), auditors must evaluate financial ratios and test journal entries for unauthorized management override. Junior auditors rely on basic manual spreadsheet filtering, which misses sophisticated posting anomalies.
- **Input Data:** Chronological general ledger transaction logs and current vs. prior period trial balances.
- **Algorithmic Workflow:**
    - Computes statutory financial ratios (Current Ratio, Quick Ratio, Gross Margin, Receivables Turnover) and flags significant period-over-period percentage variances.
    - Executes Benford's Law first-digit distribution analysis across general ledger transactions to flag unnatural numeric patterns.
    - Runs an Isolation Forest anomaly detector to identify high-risk entries (e.g., round-number postings, weekend entries, unauthorized user adjustments).
- **Required Output:** Audit Risk & Anomaly Report with visual Benford's Law deviation curves and risk-ranked journal entry flags.
- **Application Layout Placement:** Audit Analytics & Fraud Risk Dashboard view.
- **Seoul Accord Characteristics:** Characteristics 2 & 4 (Depth of statistical analysis, unfamiliar fraud patterns).

Module 9: Dynamic OpenXML Excel Workpaper Pack Exporter

- **Classification:** [Support Module - Non-CCP Platform Integration]
- **Operational Pain Point:** Financial auditors and statutory authorities reject static PDF summaries or disconnected spreadsheets. They require interactive "Audit Working Paper Packs" where editing an unadjusted ledger balance in Sheet 1 dynamically recalculates adjustments in Sheet 2, updates note disclosures in Sheet 3, and refreshes the primary Balance Sheet in Sheet 4.
- **Input Data:** Validated data graph (Trial Balances, Adjusting Entries, Statutory Statements, Numbered Notes, Tax Schedules).

- **Algorithmic Workflow:**
    - Initializes an OpenXML spreadsheet generation engine.
    - Writes dynamic Excel formulas (e.g., =SUM(Notes!C14:C22) or =TrialBalance!D10-Adjustments!E10) rather than static text into total cells.
    - Applies standard accounting formatting (currency masks, single underlines for subtotals, double underlines for grand totals).
- **Required Output:** Downloadable, fully linked Microsoft Excel (.xlsx) Audit Working Paper Pack.
- **Application Layout Placement:** Global "Export Audit Pack (.xlsx)" button in the bottom dock.
- **Seoul Accord Characteristics:** Non-CCP support module providing software engineering integration and export capabilities.

4. Hardware Optimization & Technical Architecture

Compute Resource Allocation (NVIDIA RTX 3050 / RTX 3070)

Pipeline Stage	Assigned Hardware	Execution Engine / Model	Memory & Compute Footprint
Document OCR & Visual Slicing (Module 7)	**Local GPU (RTX 3050 / 3070)**	MonkeyOCRv2 (0.7B) or OvisOCR2 (0.8B) (4-bit AWQ/GPTQ)	1.5 GB to 2.2 GB VRAM (Operates on 4 GB, 6 GB, or 8 GB GPUs)
COA Taxonomy Mapping (Module 1)	**Local GPU / CPU**	all-MiniLM-L6-v2 Sentence-Transforme r	<300 MB RAM / VRAM
Linear Arithmetic Solver (Module 2)	**Host CPU**	SciPy / PuLP Linear Programming Solvers	CPU-bound (Negligible memory, <50 ms latency)
BRS & 3-Way Reconciliation (Modules 4 & 6)	**Host CPU**	NetworkX / SciPy MILP Bipartite Solvers	CPU-bound (<200 MB RAM, <1.5 s latency)

Statutory Notes & Excel Pack (Modules 3 & 9)	**Host CPU**	OpenXML SDK / openpyxl Python Engine	CPU-bound (<100 MB RAM)
Forensic Anomaly Detector (Module 8)	**Host CPU**	scikit-learn Isolation Forest + Benford distribution	CPU-bound (<150 MB RAM)

End-to-End Technology Stack

- **AI & Numerical Optimization:** Python, PyTorch, Hugging Face Transformers, BitsAndBytes (4-bit Quantization), SciPy, PuLP, NetworkX, scikit-learn
- **Backend Services & File Processing:** FastAPI, Celery, Redis (Local task queue for background job processing), OpenXML / openpyxl
- **Data Storage:** PostgreSQL (Relational schema storage, account ontologies, and immutable audit logs)
- **Desktop Application Workspace:** React / Electron (High-density 3-pane desktop workspace, Tailwind CSS, ag-Grid)

5. Desktop Application Workspace Layout

The application utilizes a high-density, 3-pane desktop layout designed for multi-monitor accounting environments:

- **Top Navigation & Telemetry Bar:** Project/Client Switcher (Client: Alpha Corp - FY2025 Audit), Local Hardware Status indicator (RTX 3070 • 2.1GB VRAM • On-Premises Active), and Global Balance Parity Indicator ($\text{Debits} \equiv \text{Credits}$).
- **Pane 1 (Left - 35% Width): Source Document Viewer:** PDF canvas rendering synchronized bounding-box overlays (Blue = Verified Text, Green = Validated Total, Red = Flagged Arithmetic Imbalance).
- **Pane 2 (Center - 45% Width): Interactive Financial Grid:** Spreadsheet-style editor displaying live financial statements with indented sub-accounts. Broken rows display inline discrepancy badges.
- **Pane 3 (Right - 20% Width): Audit Inspector & Disclosure Studio:**
  - *Top Section:* High-resolution magnified image crop of the active cell from the source scan for character verification.
  - *Bottom Section:* Connected statutory note disclosures and applicable tax rate compliance cards.
- **Bottom Action Dock:** Global balance equation status ($\Delta = \$0.00$), "Auto-Repair via Local VLM Re-scan" button, and "Export Audit Pack (.xlsx)" button.

6. Seoul Accord Complex Computing Problem (CCP) Assessment & Form Answers

Form Question Selections

- **Range of conflicting requirements:** Involves wide-ranging or conflicting technical, computing, and other issues
- **Depth of analysis required:** Has no obvious solution, and requires conceptual thinking and innovative analysis to formulate suitable abstract models
- **Depth of knowledge required:** A solution requires the use of in-depth computing or domain knowledge and an analytical approach that is based on well-founded principles
- **Familiarity of issues:** Involves infrequently-encountered issues
- **Level of problem:** Is outside problems encompassed by standards and standard practice for professional computing
- **Extent of stakeholder involvement and level of conflicting requirements:** Involves diverse groups of stakeholders with widely varying needs
- **Consequences:** Has significant consequences in a range of contexts
- **Interdependence:** Is a high-level problem possibly including many component parts or sub-problems
- **Requirement identification:** Identification of a requirement or the cause of a problem is ill-defined or unknown

Form Selection Justification

1. **Conflicting Engineering Constraints:** The platform must achieve high extraction precision (>99%) and sub-second validation latency while executing strictly on consumer GPUs (4 GB to 8 GB VRAM) to guarantee zero data exfiltration for enterprise data privacy compliance (SOX, GDPR).
2. **Depth of Analysis & Abstract Modeling:** Autoregressive vision-language models probabilistically hallucinate digits and break under multi-page table splits. The project formulates a neuro-symbolic architecture that couples visual token extraction with deterministic Linear Programming (LP) and Mixed-Integer Linear Programming (MILP) constraint solvers, modeling financial statements as directed acyclic dependency graphs (DAGs).
3. **Depth of Multi-Disciplinary Knowledge:** The solution bridges multimodal representation learning, 4-bit edge quantization, graph-based fuzzy bipartite matching for bank reconciliation, statistical discrete anomaly detection (Benford's Law and Isolation Forests under ISA 240), and statutory financial ontologies (IFRS and Companies Act 2017 Fourth/Fifth Schedules).
4. **Interdependence & Cascading Complexity:** The system is composed of tightly coupled subsystems where optical character bounding-box extraction directly dictates state-space table reconstruction, which feeds the adjusting journal entry engine, dynamically recalculating the entire financial statement and statutory note disclosure tree.
5. **Critical Consequences:** Unlike qualitative conversational tools, financial workpapers carry direct legal, tax, and statutory liability. An undetected arithmetic mismatch or incorrect withholding tax deduction leads to regulatory penalties, tax notices, and audit disqualification.

7. Two-Semester Phased Execution Roadmap

Milestone Stage	Academic Timeline	Delivered Modules	Core Deliverables & Institutional Evidence
Gate 1: Proposal Defense	Semester 1, Week 3	Proof-of-Concept Technical Spike	Working spike demonstrating OCR table parsing and symbolic parity checking on trial balance samples; approved Scope of Work; initial RAID register.
Gates 2 & 3: Mid-1 Review	Semester 1, Week 8	Modules 1, 2, 7 (Ingestion, COA, Adjustments)	Approved SRS and SDS; fine-tuning pipeline for lightweight vision model; clickable 3-pane desktop UI prototype.
Gate 4: Final-1 Exit Gate	Semester 1, Week 16	**Committed Core System End-to-End**: Modules 1, 2, 3, 7, 9	Full working pipeline: Ingest PDF $\rightarrow$ Parse Table $\rightarrow$ Adjust Entries $\rightarrow$ Synthesize Statements + Notes $\rightarrow$ Export Dynamic Excel. Core build sign-off.
Gate 5: Mid-2 Review	Semester 2, Week 8	Modules 4, 5, 6 (BRS, Tax, 3-Way Match)	Automated Bank Reconciliation, FBR Withholding Tax validation, and 3-Way Invoice matching engines integrated into the workspace.
Gate 6: Final-2 Defense	Semester 2, Week 16	Complete Suite (All 9 Modules)	Fully containerized local desktop application; ISA 520/240 anomaly detection; complete User Acceptance Testing (UAT) report with practicing accountants.

8. Team Ownership Division (3 Members)

- **Member 1: Vision Perception & Ingestion Lead**
  - *Primary Ownership:* Module 7 (Lightweight Edge OCR) and Module 1 (COA Taxonomy Mapping).
  - *Core Responsibilities:* Document pre-processing, image deskewing, patch tiling, deploying quantized vision models (MonkeyOCRv2/OvisOCR2), token-to-pixel coordinate extraction, and Chart of Accounts semantic embedding alignment.
- **Member 2: Neuro-Symbolic Logic & Reconciliation Lead**
  - *Primary Ownership:* Module 2 (Adjusting Journal Entries), Module 4 (Bank Reconciliation), and Module 6 (3-Way Matching).
  - *Core Responsibilities:* Linear programming constraint solvers, accounting DAG dependency modeling, MILP bipartite graph matching for BRS and 3-way invoice reconciliation.
- **Member 3: Statutory Reporting, Compliance & UI Lead**
  - *Primary Ownership:* Module 3 (Statutory Statements & Notes), Module 5 (Tax Matrix), Module 8 (Audit Analytics), and Module 9 (Excel Exporter).
  - *Core Responsibilities:* Companies Act 2017 Fourth/Fifth Schedule note generator, FBR tax rule engine, Benford's Law and Isolation Forest anomaly detectors, dynamic OpenXML formula compiler, and the 3-pane desktop workspace UI.

9. Supervisor FAQ & Panel Defense Cheat-Sheet

- **Q: Why is converting a Trial Balance to a Balance Sheet considered a Complex Computing Problem?**
  - *A:* Converting a clean single-page spreadsheet is trivial. Real-world statutory accounting involves **unadjusted multi-page ledgers, cross-page continuations, period-end adjusting journal entries (AJEs), and statutory disclosure note synthesis under the Companies Act 2017**. Our computing core solves this through a coupled neuro-symbolic architecture combining visual layout decoders with linear constraint solvers and graph-based reconciliation engines.
- **Q: Why avoid cloud AI APIs (e.g., OpenAI, Google Document AI)?**
  - *A:* Financial data is legally sensitive (governed by corporate confidentiality and data protection laws). Sending unredacted enterprise ledgers to public cloud APIs violates privacy mandates. Furthermore, relying on third-party APIs violates institutional capstone ownership guidelines (Core-Replacement anti-pattern). We run compact, 4-bit quantized open-weight vision models locally on consumer GPUs.
- **Q: How does the system prevent AI hallucinations in financial figures?**
  - *A:* The AI vision model is used solely for visual perception (character recognition and layout parsing). All accounting totals, adjustments, and reconciliations are computed deterministically using linear equality solvers. If an extracted number creates an arithmetic imbalance ($\sum \text{Debits} \ne \sum \text{Credits}$), the system isolates the exact cell and crops the source image for character verification.

- **Q: What is our concrete research and engineering delta?**
    - *A:* While modern multimodal OCR models (dots.mocr, MonkeyOCRv2) extract tables as visual code representations, they lack domain-specific accounting logic, cross-page state tracking, and mathematical guarantees. Our contribution is an **audit-grade neuro-symbolic framework** that pairs lightweight edge vision models with deterministic linear constraint verification and statutory reporting compilers.